

Software Testing Documentation: Comprehensive Functional & Security Test Suite

Hello! This PDF document contains a comprehensive sample of test cases **Swag Labs E-Commerce Testing** along with real examples and screenshots from Jira boards and card structures. Please note that for some test cases, the full content is demonstrated visually through screenshots of the actual Jira cards to showcase the complete workflow and bug reporting process."

Example for Board list view on jira:

The screenshot displays a Jira Kanban board with two main columns: 'In Progress' and 'In Review'. The 'In Progress' column has a count of 29 and contains three test case cards. The 'In Review' column has a count of 0 and contains one card. The 'Done' column has a count of 0 and is empty. Each card includes a title, a description, a status icon, a key, and a priority icon.

Column	Count	Card Title	Description	Status Icon	Key	Priority Icon
In Progress	29	Test Case 1: Age Validation (Equivalence Partitioning & Boundary Value Analysis)		🚫	KAN-6	MM
		Test Case 2: Password Length Validation (Equivalence Partitioning & Boundary Value Analysis)		🚫	KAN-7	MM
		Test Case 3: API - Verify Submitting Form with Valid Fields				
In Review	0					
Done	0					

Test Case 1: Age Validation (Equivalence Partitioning & Boundary Value Analysis)

Screenshots:

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KAN-6

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Boundary Value Analysis (BVA):

Condition	Test Value	Valid / Invalid	Expected Result
Below minimum - 1	17	Invalid	Reject
Minimum value	18	Valid	Accept
Above minimum + 1	19	Valid	Accept
Below maximum - 1	59	Valid	Accept
Maximum value	60	Valid	Accept
Above maximum + 1	61	Invalid	Reject

In Progress

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Improve Bug

Details

Assignee

Unassigned

Assign to me

Parent

None

Priority

Medium

Test Case 2: Password Length Validation (Equivalence Partitioning & Boundary Value Analysis)

Screenshots:

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Test Case 2: Password Length Validation (Equivalence Partitioning & Boundary Value Analysis)

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Description

Business Rule: Password must be 8-16 characters long.

Equivalence Partitioning (EP):

Range	Test Value	Valid / Invalid	Expected Result
Password < 8 characters	5chars	Invalid	Show error: "Password must be between 8 and 16 characters"

In Progress

Improve Bug

Details

Assignee

MM Mariam Mohamed

Parent

None

Priority

Medium

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Equivalence Partitioning (EP):

Range	Test Value	Valid / Invalid	Expected Result
Password < 8 characters	5chars	Invalid	Show error: "Password must be between 8 and 16 characters"
8 <= Password <= 16 characters	Abc12345	Valid	Accept password and proceed
Password > 16 characters	VeryLongPassword12345	Invalid	Show error: "Password must be between 8 and 16 characters"

Boundary Value Analysis (BVA):

In Progress

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Details

Assignee

MM Mariam Mohamed

Parent

None

Priority

Medium

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Boundary Value Analysis (BVA):

Condition	Test Value	Valid / Invalid	Expected Result
Below minimum - 1	7 characters	Invalid	Reject
Minimum value	8 characters	Valid	Accept
Above minimum + 1	9 characters	Valid	Accept
Below maximum - 1	15 characters	Valid	Accept
Maximum value	16 characters	Valid	Accept
Above maximum + 1	17 characters	Invalid	Reject

In Progress

Improve Bug

Details

Assignee

MM Mariam Mohamed

Parent

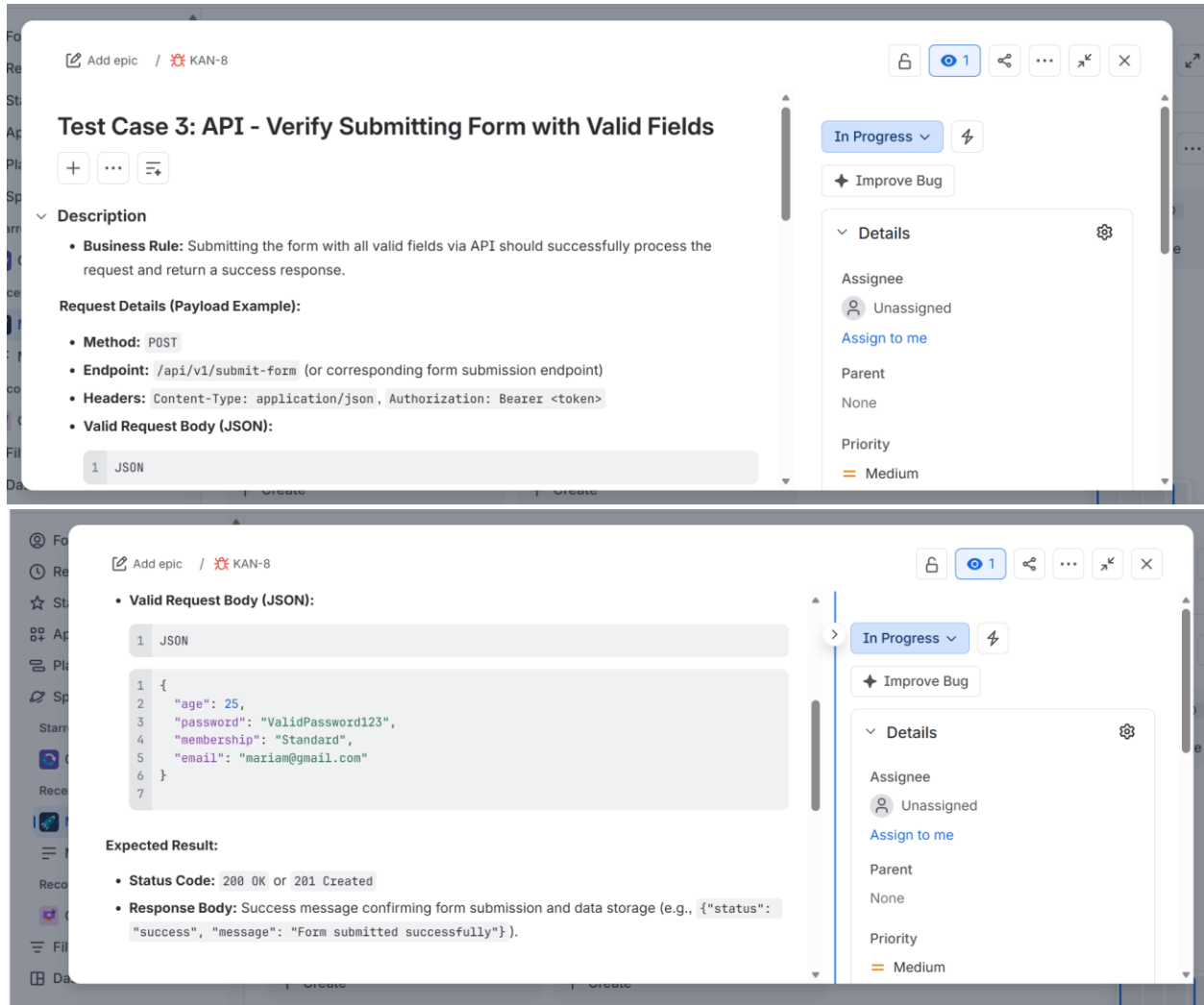
None

Priority

Medium

Test Case 3: API - Verify Submitting Form with Valid Fields

Screenshots:



Test Case 4: API - Verify Submitting Form with Missing Name Field

Content:

- Business Rule:** Submitting the form with a missing required field (name) via API should reject the request and return a validation error.

Request Details (Payload Example):

- Method:** POST
- Endpoint:** /api/v1/submit-form
- Headers:** Content-Type: application/json, Authorization: Bearer <token>

- **Invalid Request Body (JSON - Missing name):**

JSON

```
{  
  "age": 25,  
  "password": "ValidPassword123",  
  "membership": "Standard",  
  "email": "mariam@gmail.com"  
}
```

Expected Result:

- **Status Code:** 400 Bad Request or 422 Unprocessable Entity
- **Response Body:** Error message indicating the required field is missing (e.g., {"error": "The name field is required."}).

Test Case 5: API - Verify Submitting Form with Student Status

Content:

- **Business Rule:** Submitting the form with the student status field correctly populated via API should process the student eligibility and apply appropriate rules or pricing.

Request Details (Payload Example):

- **Method:** POST
- **Endpoint:** /api/v1/submit-form
- **Headers:** Content-Type: application/json, Authorization: Bearer <token>
- **Valid Request Body (JSON - Including is_student):**

JSON

```
{  
  "name": "Mariam Mohamed",  
  "age": 21,  
  "password": "ValidPassword123",  
  "membership": "Standard",  
  "email": "mariam@gmail.com",  
  "is_student": true  
}
```

}

Expected Result:

- **Status Code:** 200 OK or 201 Created
- **Response Body:** Success confirmation reflecting that the student status has been successfully captured and applied (e.g., {"status": "success", "is_student": true, "message": "Form submitted successfully"}).

Test Case 6: Decision Table - Promo Code & Membership Validation

Content:

- **Business Rule:**
 - If **Premium** is selected with a valid promo code, the user gets a **20% discount**.
 - If **Premium** is selected without a promo code (or invalid code), the user pays the **full price**.
 - If **Standard** is selected, any promo code is ignored and the user pays the **full price**.

Decision Table Format (for Jira description):

Membership Type	Promo Code	Expected Result
Premium	Valid	20% discount
Premium	Invalid	Pay the Full Price
Standard	Valid / Invalid	Promo ignored $\rightarrow$ Pay the Full Price

Test Case 7: Login Screen - Failed Login Attempts & Account Lockout

Content:

- **Business Rule:** After **3 consecutive failed login attempts**, the account is locked for **15 minutes**.

Equivalence Partitioning (EP):

Failed Login Attempts	Valid / Invalid	Expected Result
attempts < 3	Valid	Login retry allowed
attempts >= 3	Invalid	Account locked for 15 minutes

Boundary Value Analysis (BVA):

Condition	Test Value	Valid / Invalid	Expected Result
Below limit (Allowed retry)	2 attempts	Valid	Login retry allowed
Exact limit (Trigger lockout)	3 attempts	Invalid	Account locked for 15 minutes
Above limit (Locked out)	4 attempts	Invalid	Account locked for 15 minutes

.

Test Case 8: Decision Table Testing (General Form Rules)

Content:

- **Business Rule:** Test various combinations of user inputs (such as Age, Password length, and Membership tier) to ensure form validation handles multiple conditional rules correctly.

Decision Table Format (for Jira description):

Age	Password	Membership	Expected Result
18	7 char	Standard	Invalid (Password too short)
17	8 char	Standard	Invalid (Age below minimum)
60	9 char	Premium	Valid (All fields match criteria)
35	16 char	Premium	Valid (All fields match criteria)
20	17	Standard	Invalid (Password too long)

Test Case 9: Negative Test Case - Invalid Email Format

Content:

- - **Business Rule:** The email field must contain a valid format with an @ symbol and a domain (e.g., .com).
 - **Equivalence Partitioning (EP):**
 - Email without '@' or domain $\rightarrow$ **Invalid** (Format is incorrect, system rejects it).
 - **Test Values Table:**

Email Input	Valid / Invalid	Expected Result
mariamgmail.com	Invalid	Show error: "Invalid email format (missing @)"
mariam@	Invalid	Show error: "Invalid email format (missing domain)"

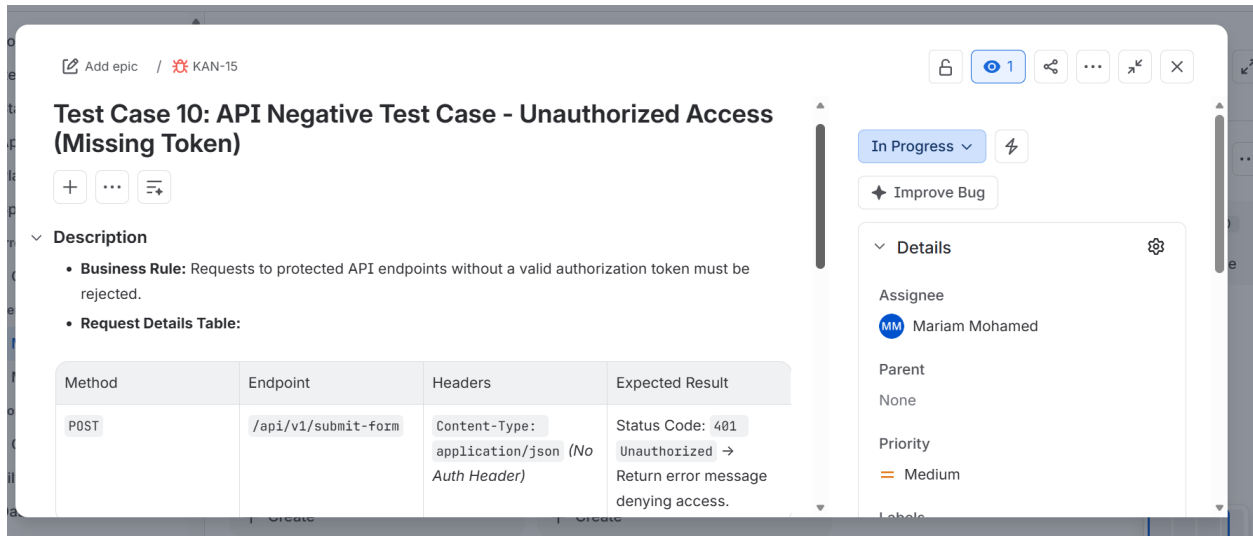
•

Test Case 10: API Negative Test Case - Unauthorized Access (Missing Token)

Content:

- - **Business Rule:** Requests to protected API endpoints without a valid authorization token must be rejected.
 - **Request Details Table:**

Method	Endpoint	Headers	Expected Result
POST	/api/v1/submit-form	Content-Type: application/json (No Auth Header)	Status Code: 401 Unauthorized $\rightarrow$ Return error message denying access.



Test Case 11: UI/UX Field Validation - Required Fields

Content:

- Business Rule:** Mandatory form fields (such as Name, Email, and Password) must not be left empty, and the system should display validation errors if a user attempts to submit a blank form.
 - Test Values Table:**

Field Name	Input Value	Valid / Invalid	Expected Result
Name	(Empty)	Invalid	Show error: "Name is required"
Email	(Empty)	Invalid	Show error: "Email is required"
Password	(Empty)	Invalid	Show error: "Password is required"

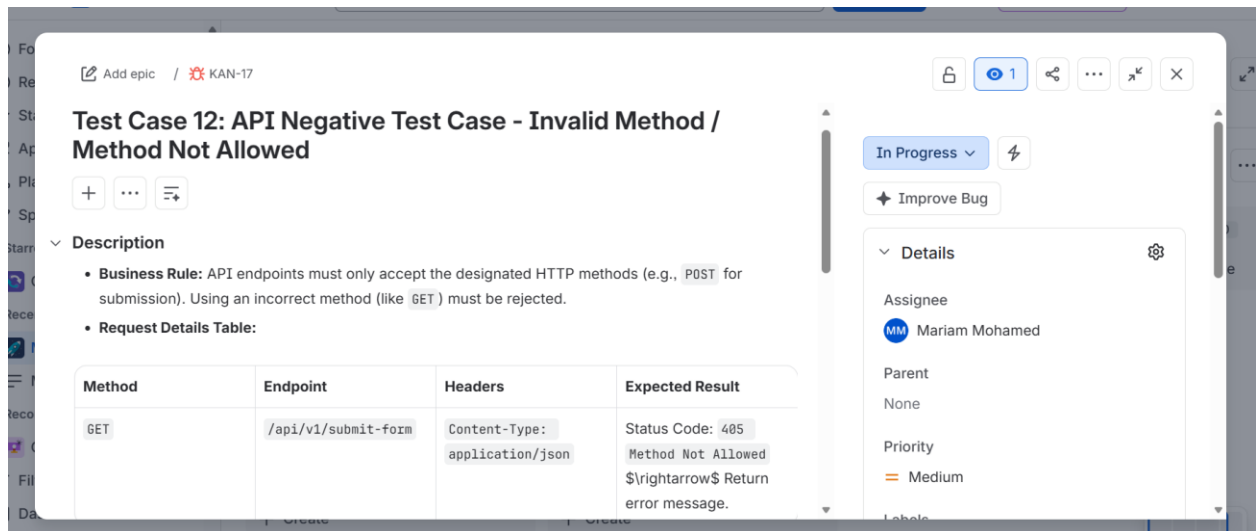
Test Case 12: API Negative Test Case - Invalid Method / Method Not Allowed

Content:

- Business Rule:** API endpoints must only accept the designated HTTP methods (e.g., POST for submission). Using an incorrect method (like GET) must be rejected.
 - Request Details Table:**

Method	Endpoint	Headers	Expected Result
GET	/api/v1/submit-form	Content-Type: application/json	Status Code: 405 Method Not Allowed \$rightarrow\$ Return error message.

Screenshot:



Test Case 13: Boundary Value Analysis - Maximum Age Limit

Content:

- **Business Rule:** The maximum allowed age for registration is 60 years. Testing boundary values near the maximum limit to ensure accurate edge-case handling.
- **Test Values Table:**

Age Input	Valid / Invalid	Expected Result
59	Valid	Accept age (Below maximum - 1)
60	Valid	Accept age (Exact maximum boundary)
61	Invalid	Reject age (Above maximum + 1)

Test Case 14: Password Character Complexity Validation

Content:

- **Business Rule:** The password must contain a mix of uppercase letters, lowercase letters, numbers, and special characters to ensure robust security.

- **Test Values Table:**

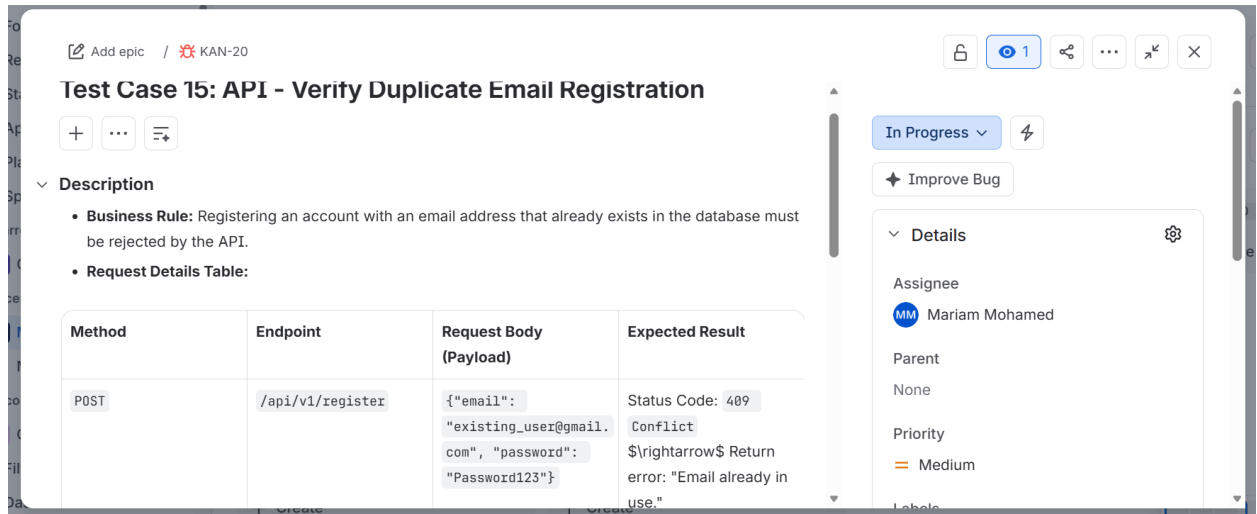
Password Input	Valid / Invalid	Expected Result
abcdefgh	Invalid	Reject password (Missing numbers and special characters)
ABCDEFGH	Invalid	Reject password (Missing lowercase letters and numbers)
Abcdef1!	Valid	Accept password (Meets all complexity criteria)

Test Case 15: API - Verify Duplicate Email Registration

Content:

- - **Business Rule:** Registering an account with an email address that already exists in the database must be rejected by the API.
 - **Request Details Table:**

Method	Endpoint	Request Body (Payload)	Expected Result
POST	/api/v1/register	<pre>{ "email": "existing_user@gmail.com", "password": "Password123" }</pre>	Status Code: 409 Conflict ↳ Return error: "Email already in use."



Test Case 16: API - Update User Profile Data

Content:

- Business Rule:** Authorized users can update their profile information (such as name or membership status) via the API by sending a valid PUT or PATCH request.
 - Request Details Table:**

Method	Endpoint	Headers	Request Body (Payload)	Expected Result
PUT	/api/v1/user/profile	Authorization: Bearer <token>, Content-Type: application/json	{ "name": "Mariam Mohamed Updated", "membership": "Premium"} }	Status Code: 200 OK → Return updated profile data and success message.

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In Progress

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Improve Bug

Details

Assignee

Unassigned

Assign to me

Parent

None

Priority

Medium

Method

Endpoint

Headers

Request Body (Payload)

Expected Result

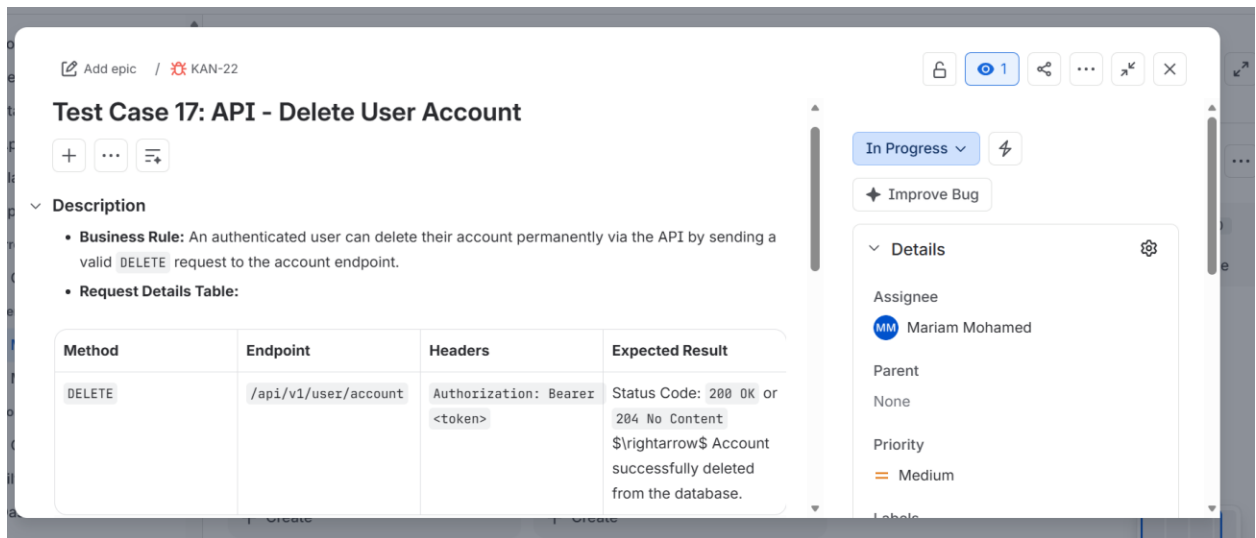
PUT	/api/v1/user/profile	Authorization: Bearer <token>, Content-Type: application/json	<pre>{ "name": "Mariam Mohamed", "membership": "Premium" }</pre>	Status Code: 200 OK → Return updated profile data and success message.
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Test Case 17: API - Delete User Account

Content:

- Business Rule:** An authenticated user can delete their account permanently via the API by sending a valid DELETE request to the account endpoint.
 - Request Details Table:**

Method	Endpoint	Headers	Expected Result
DELETE	/api/v1/user/account	Authorization: Bearer <token>	Status Code: 200 OK or 204 No Content → Account successfully deleted from the database.

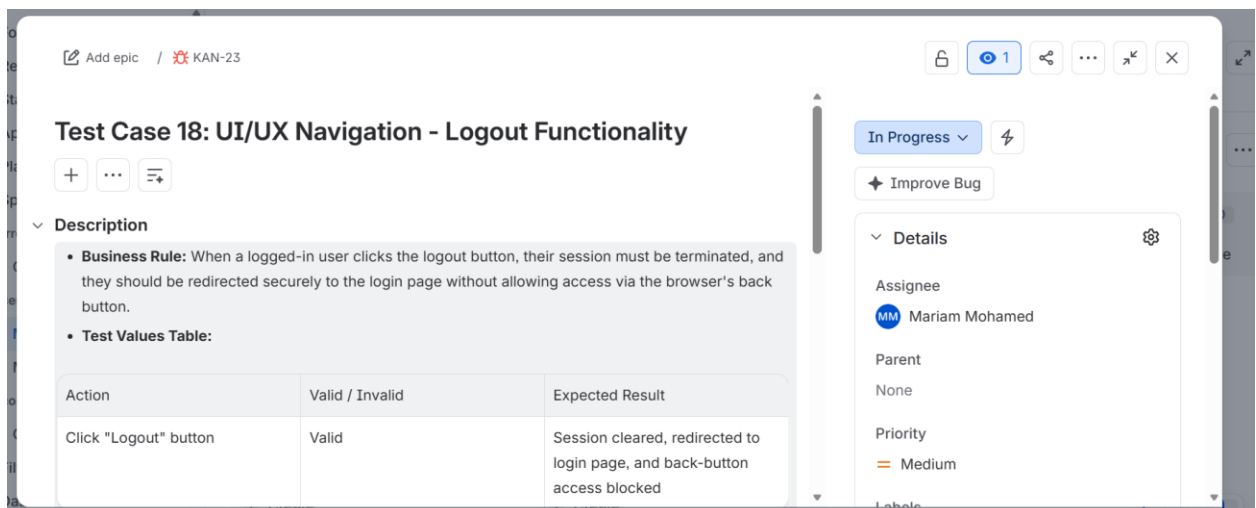


Test Case 18: UI/UX Navigation - Logout Functionality

Content:

- Business Rule:** When a logged-in user clicks the logout button, their session must be terminated, and they should be redirected securely to the login page without allowing access via the browser's back button.
 - Test Values Table:**

Action	Valid / Invalid	Expected Result
Click "Logout" button	Valid	Session cleared, redirected to login page, and back-button access blocked



Test Case 19: UI/UX - Remember Me Functionality

Content:

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- **Business Rule:** When a user checks the "Remember Me" box during login, their session token or authentication state should persist across browser restarts.
- **Test Values Table:**

Action	Valid / Invalid	Expected Result
Login with "Remember Me" checked, then close and reopen browser	Valid	User remains logged in without re-entering credentials

Test Case 20: API - Rate Limiting (Too Many Requests)

Content:

- **Business Rule:** To prevent brute-force attacks, the API must limit the number of rapid consecutive login attempts from the same IP address or user account.
- **Request Details Table:**

Method	Endpoint	Action / Frequency	Expected Result
POST	/api/v1/login	Send 10 rapid requests within 5 seconds with incorrect credentials	Status Code: 429 Too Many Requests → Block further requests temporarily and return error message.

Test Case 20: API - Rate Limiting (Too Many Requests)

Description

- Business Rule:** To prevent brute-force attacks, the API must limit the number of rapid consecutive login attempts from the same IP address or user account.
- Request Details Table:**

Method	Endpoint	Action / Frequency	Expected Result
POST	/api/v1/login	Send 10 rapid requests within 5 seconds with incorrect credentials	Status Code: 429 Too Many Requests → Block further requests temporarily and return error message.

Details

Assignee: MM Mariam Mohamed

Parent: None

Priority: Medium

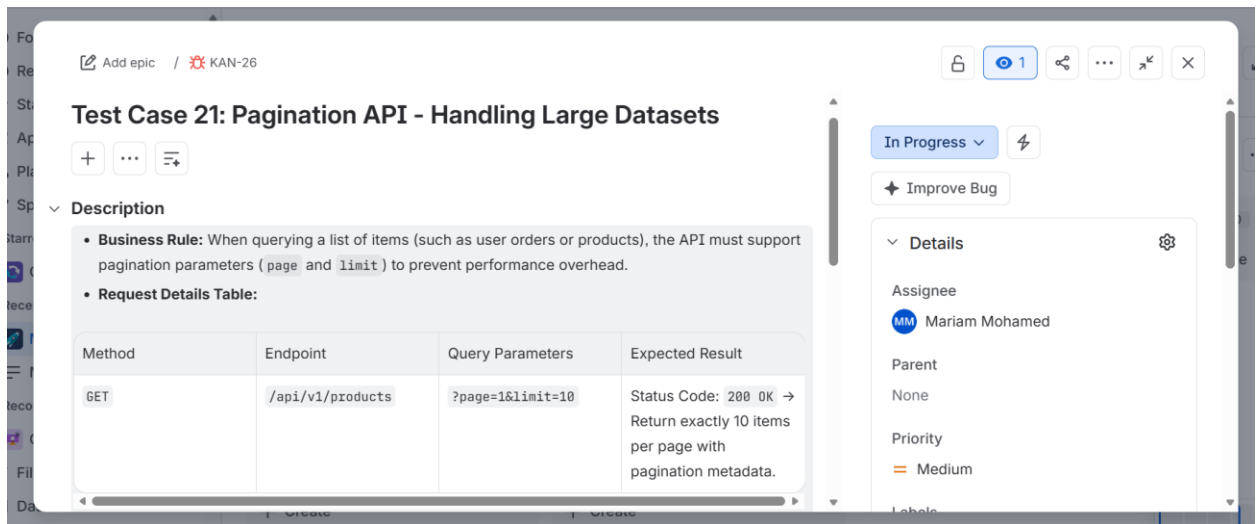
Test Case 21: Pagination API - Handling Large Datasets

Content:

-

- **Business Rule:** When querying a list of items (such as user orders or products), the API must support pagination parameters (page and limit) to prevent performance overhead.
- **Request Details Table:**

Method	Endpoint	Query Parameters	Expected Result
GET	/api/v1/products	?page=1&limit=10	Status Code: 200 OK → Return exactly 10 items per page with pagination metadata.



Test Case 22: File Upload API - Format and Size Validation

Content:

- **Business Rule:** When uploading user profile pictures or attachments via the API, the system must accept only allowed formats (e.g., .png, .jpg) and enforce a maximum file size limit (e.g., 5MB).
- **Request Details Table:**

Method	Endpoint	Payload / File Type	Expected Result
POST	/api/v1/upload	File: document.pdf (Invalid format) or large_image.png (> 5MB)	Status Code: 400 Bad Request → Return error: "Invalid file type or size exceeds limit."

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Test Case 22: File Upload API - Format and Size Validation

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Description

- Business Rule:** When uploading user profile pictures or attachments via the API, the system must accept only allowed formats (e.g., .png, .jpg) and enforce a maximum file size limit (e.g., 5MB).
- Request Details Table:**

Method	Endpoint	Payload / File Type	Expected Result
POST	/api/v1/upload	File: document.pdf (Invalid format) or large_image.png (> 5MB)	Status Code: 400 Bad Request → Return error: "Invalid file type or size exceeds limit."

In Progress

Improve Bug

Details

Assignee
Unassigned
Assign to me

Parent
None

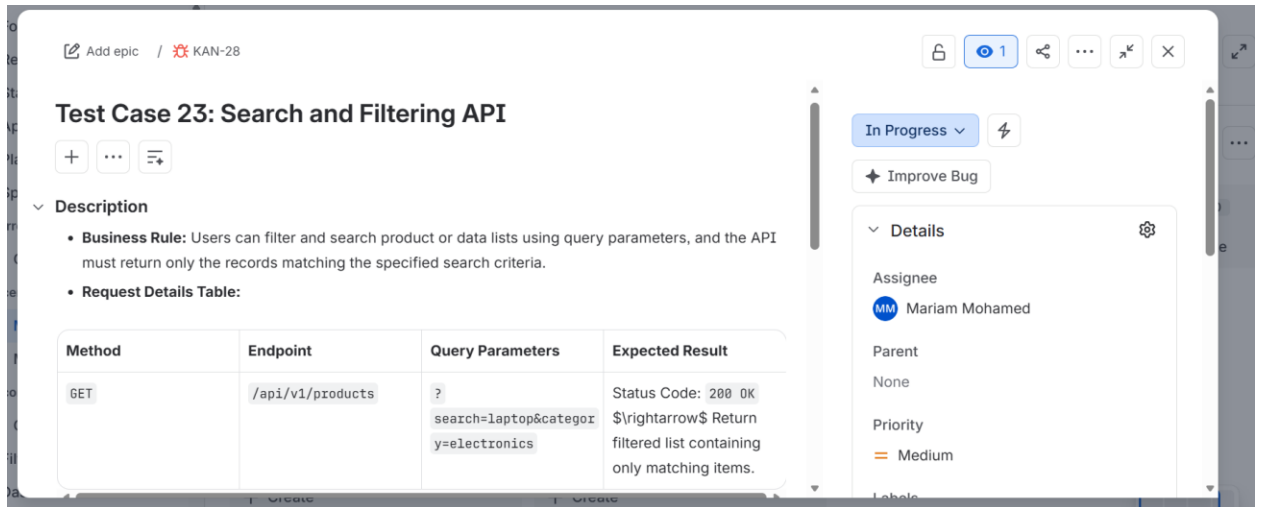
Priority
Medium

Test Case 23: Search and Filtering API

Content:

- Business Rule:** Users can filter and search product or data lists using query parameters, and the API must return only the records matching the specified search criteria.
 - Request Details Table:**

Method	Endpoint	Query Parameters	Expected Result
GET	/api/v1/products	?search=laptop&category=electronics	Status Code: 200 OK → Return filtered list containing only matching items.



Test Case 24: Password Reset - Token Expiration

Content:

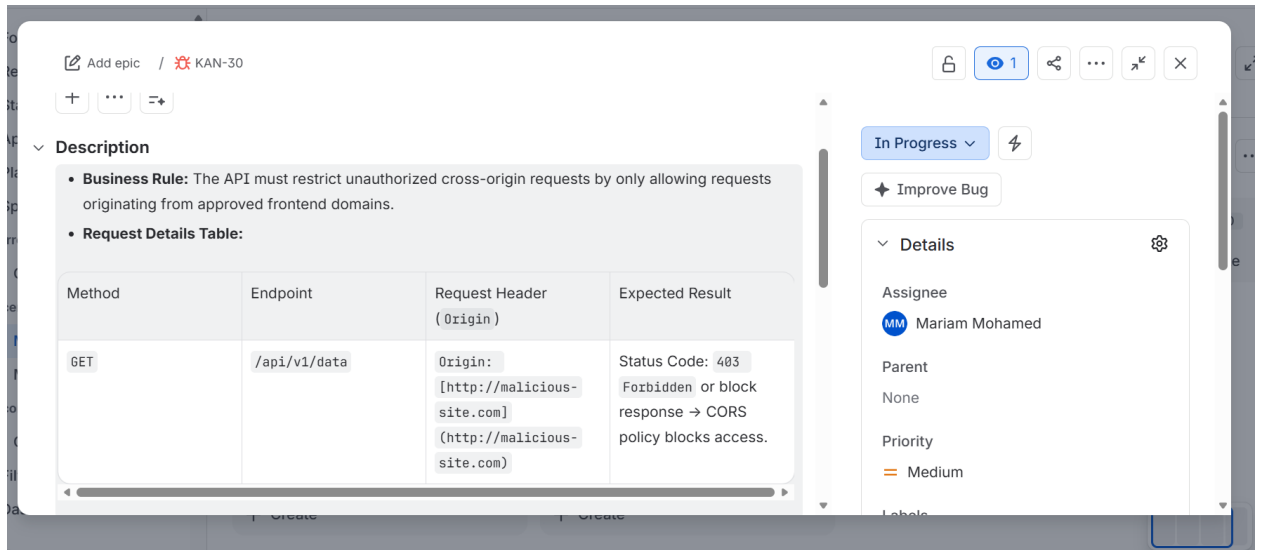
- Business Rule:** Password reset links sent via email must expire after a specific time frame (e.g., 30 minutes) to prevent unauthorized access.
 - Test Values Table:**

Action	Valid / Invalid	Expected Result
Click reset link after expiration time	Invalid	Show error: "Password reset token has expired. Please request a new one."

Test Case 25: Cross-Origin Resource Sharing (CORS) Configuration

- Business Rule:** The API must restrict unauthorized cross-origin requests by only allowing requests originating from approved frontend domains.
 - Request Details Table:**

Method	Endpoint	Request Header (Origin)	Expected Result
GET	/api/v1/data	Origin: http://malicious-site.com	Status Code: 403 Forbidden or block response → CORS policy blocks access.

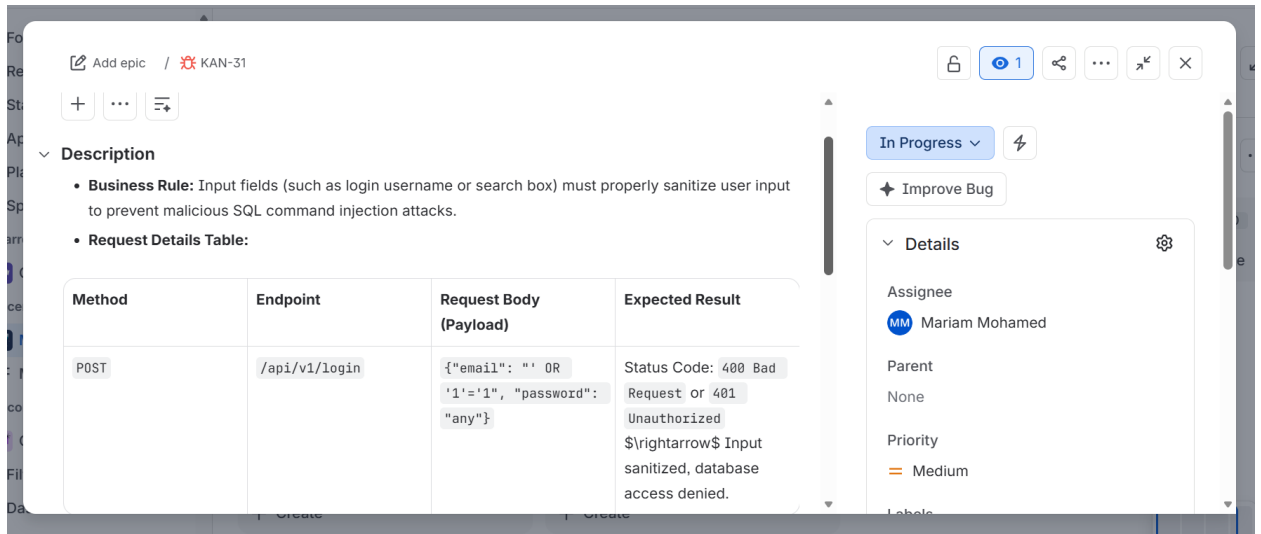


Test Case 26: Security - SQL Injection Prevention

Content:

- **Business Rule:** Input fields (such as login username or search box) must properly sanitize user input to prevent malicious SQL command injection attacks.
 - **Request Details Table:**

Method	Endpoint	Request Body (Payload)	Expected Result
POST	/api/v1/login	{ "email": " ' OR '1'='1", "password": "any" }	Status Code: 400 Bad Request or 401 Unauthorized \$→\$ Input sanitized, database access denied.

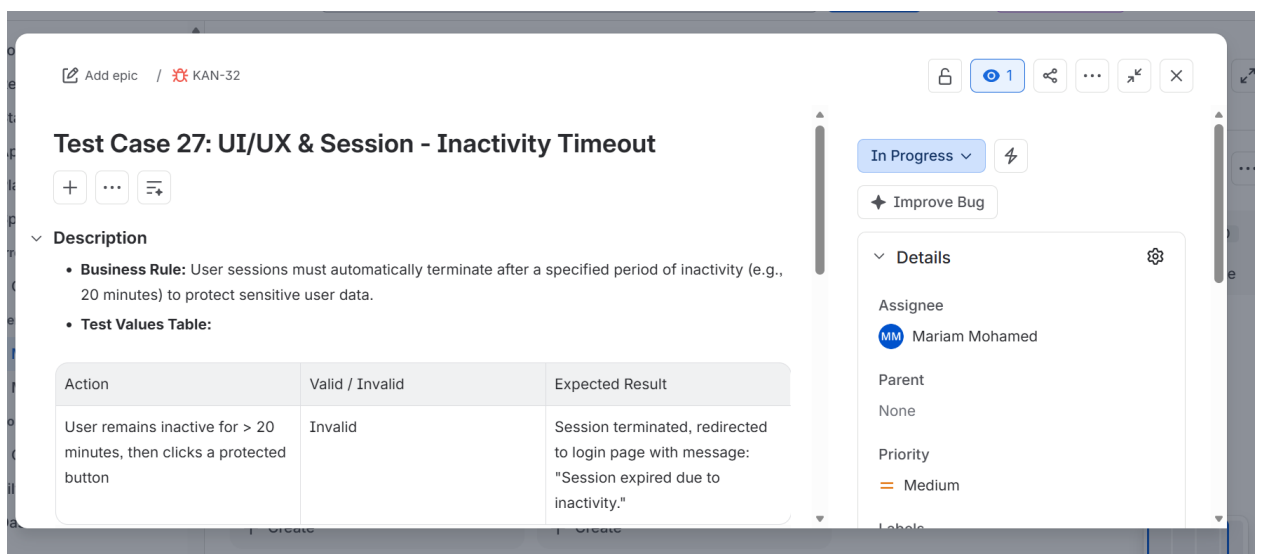


Test Case 27 UI/UX & Session - Inactivity Timeout

Content:

- **Business Rule:** User sessions must automatically terminate after a specified period of inactivity (e.g., 20 minutes) to protect sensitive user data.
 - **Test Values Table:**

Action	Valid / Invalid	Expected Result
User remains inactive for > 20 minutes, then clicks a protected button	Invalid	Session terminated, redirected to login page with message: "Session expired due to inactivity."

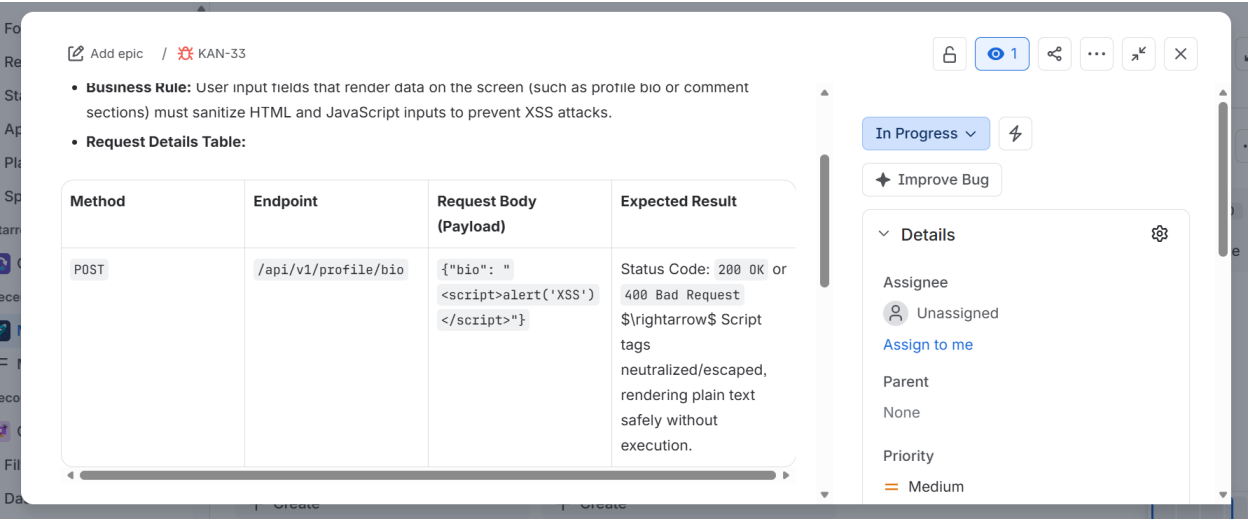


Test Case 28: Security - Cross-Site Scripting (XSS) Prevention

Content:

- Business Rule:** User input fields that render data on the screen (such as profile bio or comment sections) must sanitize HTML and JavaScript inputs to prevent XSS attacks.
 - Request Details Table:**

Meth od	Endpoint	Request Body (Payload)	Expected Result
POST	/api/v1/profile /bio	<pre>{"bio": " <script>alert('XSS')</sc ript>"}</pre>	Status Code: 200 OK or 400 Bad Request \$rightarrow\$ Sc ript tags neutralized/esca ped, rendering plain text safely without execution.



Test Case 29: Localization and Multi-Language Support

Content:

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- **Business Rule:** The application must correctly switch UI labels, messages, and layout direction (e.g., LTR for English, RTL for Arabic) based on the user's selected language preference.
- **Test Values Table:**

Action	Valid / Invalid	Expected Result
Switch language setting to Arabic	Valid	UI text translates to Arabic, and page direction changes to Right-to-Left (RTL)

The screenshot shows a Jira ticket interface for 'Test Case 29: Localization and Multi-Language Support'. The ticket is in 'In Progress' status and is assigned to 'Unassigned'. The description contains a business rule and a test values table.

Test Case 29: Localization and Multi-Language Support

Description

- Business Rule:** The application must correctly switch UI labels, messages, and layout direction (e.g., LTR for English, RTL for Arabic) based on the user's selected language preference.
- Test Values Table:**

Action	Valid / Invalid	Expected Result
Switch language setting to Arabic	Valid	UI text translates to Arabic, and page direction changes to Right-to-Left (RTL)

Details

- Assignee: Unassigned
- Parent: None
- Priority: Medium